Estimating mitragynine and 7-hydroxymitragynine exposure from self-reported kratom leaf-powder doses

Literature review, pharmacokinetic model, calibration and implementation specification for the *bezKRATOMU* application

Technical report (model version 2)

6 November 2026; version 2: 12 November 2026

Abstract

We describe how the *bezKRATOMU* tapering application will estimate the time course of the two principal opioid-active kratom alkaloids, mitragynine (MG) and its metabolite 7-hydroxymitragynine (7-OH), from the doses (grams and time) that users record. Human pharmacokinetic (PK) studies published 2015–2026 are reviewed. Oral MG is absorbed rapidly ($t_{\max} \approx$ 1–1.5 h) and declines bi-exponentially, with a fast distribution phase over the first ∼6 h and a long terminal phase (reported half-lives 9–68 h depending on dose and sampling duration). 7-OH is formed from MG mainly by hepatic CYP3A4, peaks slightly later ($t_{\max} \approx$ 1.5–2 h) and tracks the parent at a concentration ratio of about 0.1–0.3. We adopt a published linked parent–metabolite two-compartment model (Notario *et al.*, 2026), recalibrate it to the largest dose-ranging study of encapsulated leaf powder (Huestis *et al.*, 2024) with two scale factors ($F_{\mathrm{rel}} = 0.277$, $s_{\mathrm{OH}} = 1.85$), and validate it against independent steady-state data after 15 daily doses. The two analytes are combined into a single potency-weighted index of μ-opioid receptor activity, which is displayed on a relative scale (100 % = the steady-state daily mean at the user's baseline daily amount). Because the model is linear, the curve for any dosing history is an exact superposition of a five-term sum of exponentials, whose coefficients are tabulated here. The result is a population-average estimate, not a measurement: between-subject variability in the source studies is 5–10-fold.

Version 2 (§9) asks whether the model holds for the app's actual users, who have taken kratom daily for 1–4 years, some at more than 30 g/day. A systematic search (PubMed, Europe PMC full text, OpenAlex citation tracking, preprint servers, ClinicalTrials.gov) found no time-resolved PK data in long-term heavy users. Data from regular users at typical doses (three studies) and single doses up to 12 g (one study) show that the model's *shape* and linearity hold; only its absolute exposure is about 2× too low under real-world (fed) conditions, a factor that cancels out of every quantity the app displays. The central curve is therefore unchanged. What remains uncertain is how fast the level falls, so the app now shows a range for the projection (terminal half-life 20–60 h, extended to 84 h for users above 12 g/day).

Contents

1 Background pharmacology **2**

2 Human pharmacokinetic literature **3**
2.1 Absorption . . . 3
2.2 Distribution and elimination . . . 3
2.3 Linearity and variability . . . 4
2.4 Formulation . . . 4

3 Model **4**
3.1 Structure . . . 4
3.2 Direct 7-OH in the powder . . . 5
3.3 Analytical solution and superposition . . . 5

4 Calibration and validation **5**
4.1 Procedure . . . 5
4.2 Validation against repeated dosing . . . 6

5 A combined activity index **7**

6 Relative scale shown to the user **7**

7 Amount remaining in the body **7**

8 Model behaviour **8**

9 Regular and heavy users (version 2) **9**
9.1 Question . . . 9
9.2 Search . . . 9
9.3 What exists . . . 10
9.4 Comparison with the model . . . 10
9.5 Range shown for the decline . . . 12

10 Uncertainty and limitations **13**

11 Implementation specification **13**

Scope and caveat. The model estimates what an *average* healthy adult's plasma concentrations would be after the logged doses of a leaf powder containing 1.6 % MG. It does not measure anything, it does not predict effects, craving or withdrawal, and it must not be used to decide when or how much to dose. It is not valid for kratom extracts or 7-OH products.

1 Background pharmacology

Alkaloids. Dried *Mitragyna speciosa* leaf contains dozens of indole and oxindole alkaloids. Mitragynine is the most abundant (about 1–2 % of leaf mass; 1.33 % in the product of Huestis *et al.* [1]; 1.6 % in the product used by our users). 7-Hydroxymitragynine is present only in traces in leaf (<0.01 % in [1]; 0.001 % in our users' product). Other alkaloids (e.g. speciociliatine, paynantheine, speciogynine) reach substantial plasma levels [3] but are less characterised and are not modelled here.

Metabolic origin of 7-OH. In humans, 7-OH plasma exposure after kratom is predominantly the result of hepatic oxidation of MG by CYP3A4. Pretreatment with the selective CYP3A4 inhibitor itraconazole reduced 7-OH $C_{\max}$ by 56 % and $\mathrm{AUC}_{0\text{–}72}$ by 43 %, while MG $C_{\max}$ rose about 1.5-fold [4]. A physiologically based model reached the same conclusion (CYP3A4 inhibitors reduce 7-OH formation but change MG exposure little) [7]. In mice, 7-OH formed from MG mediates a large part of MG's analgesic effect [9].

Receptor pharmacology. Both compounds are G-protein-biased partial agonists at the human μ-opioid receptor (MOR). In a BRET G-protein activation assay, MG had $EC_{50} = 339 \pm 178$ nM ($E_{\max} = 34$ %) and 7-OH $EC_{50} = 34.5 \pm 4.5$ nM ($E_{\max} = 47$ %) [8, 10], i.e. 7-OH is about tenfold more potent. Binding affinities differ between assays (7-OH K_i 8–47 nM; MG $K_i \approx 230$ nM) [10]. MG also acts at adrenergic and other receptors; these non-opioid actions are ignored in this report.

Protein binding. Both MG and 7-OH are more than 90 % bound in human plasma (equilibrium dialysis) [22]; reported MG binding is 85–97 %. As the unbound fractions are similar, we compare total concentrations.

2 Human pharmacokinetic literature

Table 1: Human PK studies used in this report. Values are means or medians as reported by the authors. MG = mitragynine; 7-OH = 7-hydroxymitragynine; SD = single dose; MD = multiple (daily) doses.

Study	Population and dose	Mitragynine	7-OH
Trakulsrichai 2015 [5]	9 chronic male users; tea, 6.25–23 mg MG; no washout	t_{max} 0.83 h; $t_{1/2}$ 23.2±16.1 h; bi-exponential; C_{max}, AUC proportional to dose	not measured
Tanna 2022 [3]	5–6 healthy adults; 2 g leaf powder as tea	t_{max} 1 h; $t_{1/2}$ 45.3 h (median); two-compartment model fits best	t_{max} 1 h; $t_{1/2}$ 5.7 h; AUC ratio to MG 0.24
Mongar 2024 [4]	15–16 healthy adults; tea, 23.6 mg MG; ± itraconazole	C_{max} 159 ng/mL at 0.84 h; $t_{1/2}$ 8.7 h (72 h sampling); biphasic	C_{max} 12.8 ng/mL at 1.77 h; $t_{1/2}$ 7.4 h; CYP3A4-dependent
Huestis 2024 [1]	8–13 per arm, healthy adults; capsules 0.5–4 g leaf (6.65–53.2 mg MG); SD and 15 daily doses	t_{max} 1.0–1.3 h; C_{max} dose-proportional; median $t_{1/2}$ 3.7→42.9 h (SD), 14.6→61.2 h (MD); steady state 8–9 d	t_{max} 1.2–1.8 h; $t_{1/2}$ 1.7–4.7 h (SD); ratio to MG 0.2–0.3 (SD), 0.15–0.21 (MD)
Huestis 2026 [2]	Extract (39.5 % MG), 9.9–59.2 mg MG; SD and 15 MD	C_{max} 32.8–196 ng/mL; t_{max} 1.3–2 h; AUC < 1.4× proportional; higher exposure than leaf	C_{max} 6.8–30.5 ng/mL; t_{max} 1.3–2.3 h
Notario 2026 [6]	Model fitted (Bayesian, HMC) to Mongar 2024 data	linked parent–metabolite, 2+2 compartments, first-order absorption	formation fraction $F_m = 0.65$
Added in version 2 (§9)			
Reissig 2026 [11]	6 per cohort, opioid-experienced non-dependent adults; capsules 1–12 g leaf (∼1 % MG, 10–123 mg); single dose, fed	C_{max} 42–379 ng/mL; AUC_{0-48}/mg constant (23.6–31.7); $t_{1/2}$ 16–22 h (48 h sampling)	C_{max} 6.7–58 ng/mL
Smith 2024 [12]	10 regular users (6.6 y, 2–14 y); own product, usual dose 1.1–10.9 g	post-dose 176 (37–396) ng/mL	34 (3–78) ng/mL
Kanumuri 2026 [13]	4 regular users (Johns Hopkins trial); usual dose 27–54 mg MG	pre-dose 27–222 ng/mL; t_{max} 1–1.5 h; $t_{1/2}$ 17–35 h	AUC ratio to MG 0.11–0.13

2.1 Absorption

All studies agree that absorption of oral MG is fast: median t_{max} is 0.8–1.3 h after tea or capsules and 1.3–2.0 h at steady state or after extracts. 7-OH appears in plasma within 15 min [4] and peaks 0.2–0.9 h after MG, as expected for a formed metabolite. Some participants showed secondary 7-OH peaks after a meal 4 h post-dose [1]. Tanna *et al.* noted that sparse early sampling limited the characterisation of absorption [3].

2.2 Distribution and elimination

Concentration–time profiles are bi-exponential: "an initial rapid descent after C_{max} for about 6 h and then a much slower terminal phase" [1]; see also [5, 3, 4]. Apparent volumes are very large (MG V_z/F 1 000–13 000 L), consistent with extensive tissue distribution of a lipophilic base. The terminal half-life reported for MG varies widely (8.7–68 h). Most of this spread is methodological:

- *Sampling duration and assay sensitivity.* The slow phase is visible only as long as concentrations stay above the limit of quantification. Studies that sample for 72 h [4] or give low doses report short half-lives, and studies that sample for 5–10 days at higher doses report long ones [1, 3]. Huestis *et al.* state that half-lives "were best reflected at higher doses due to the ability to measure low analyte concentrations longer" [2].
- *Possible saturation.* The same authors also raise possible enzyme saturation at higher doses [2]. AUC rose slightly more than dose-proportionally (1.18–1.42-fold over the dose range) [1].

The terminal phase carries little of the total exposure: in [1] AUC_{0-24} was already 61–80 % of $AUC_{0-\infty}$ after one dose. Accumulation on once-daily dosing is therefore modest (MG C_{max} ratio 1.1–1.3, AUC ratio 1.6–1.9;

7-OH 0.9–1.0 and 1.0–1.3) [1].

2.3 Linearity and variability

$C_{\max}$ of both analytes increased dose-proportionally over 6.65–53.2 mg MG, and AUC increased slightly more than proportionally [1]; extracts gave less than 1.4× the proportional increase [2]. Trakulsrichai found linear kinetics over 6–23 mg [5]. A linear model is therefore a reasonable first approximation. Between-subject variability is large: at 0.5 g leaf, MG $C_{\max}$ ranged 1.8–61.5 ng/mL; at 4 g, 34–204 ng/mL [1]. Sources of variability include CYP3A4 activity [4, 2], food, and product composition.

2.4 Formulation

Exposure per mg of MG differs between preparations and studies: the tea of Mongar *et al.* gave $\mathrm{AUC}_{0-\infty}$ of 940 ng h/mL for 23.6 mg MG [4], whereas 26.6 mg in capsules gave 364 ng h/mL [1], and extracts gave more than leaf powder [2]. Because our users take plain leaf powder, the model is calibrated to the capsule study [1], which used the same material.

3 Model

3.1 Structure

We adopt the linked parent–metabolite model of Notario *et al.* [6], which describes both analytes with two compartments each and first-order absorption (Figure 1). Two empirical scale factors adapt it from tea to leaf powder (§4).

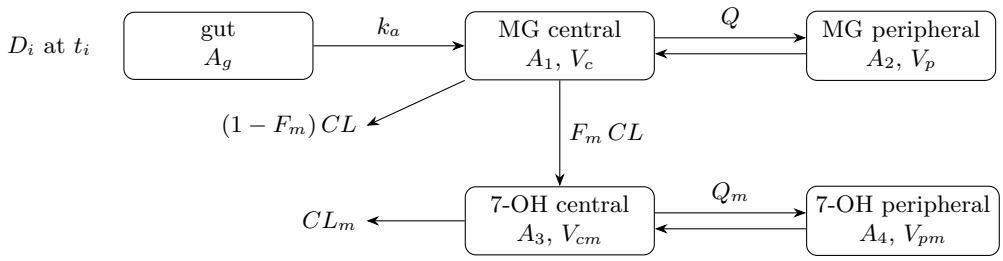


Figure 1: Model structure. Each dose D_i (mg MG) enters the gut at time t_i. A fraction F_m of MG clearance forms 7-OH. All parameters are apparent (divided by the unknown oral bioavailability).

With amounts A in μg and volumes in L (so that A/V is in ng/mL), the system is

$$\dot{A}_g = -k_a A_g, \tag{1}$$

$$\dot{A}_1 = k_a A_g - \frac{CL + Q}{V_c} A_1 + \frac{Q}{V_p} A_2, \qquad \dot{A}_2 = \frac{Q}{V_c} A_1 - \frac{Q}{V_p} A_2, \tag{2}$$

$$\dot{A}_3 = F_m \frac{CL}{V_c} A_1 - \frac{CL_m + Q_m}{V_{cm}} A_3 + \frac{Q_m}{V_{pm}} A_4, \qquad \dot{A}_4 = \frac{Q_m}{V_{cm}} A_3 - \frac{Q_m}{V_{pm}} A_4, \tag{3}$$

with plasma concentrations $C_{\mathrm{MG}} = A_1/V_c$ and $C_{\mathrm{OH}} = s_{\mathrm{OH}}\, A_3/V_{cm}$. A dose of g grams of powder contains $D = 1000\, f_{\mathrm{MG}}\, g$ mg MG ($f_{\mathrm{MG}} = 0.016$, i.e. 16 mg per gram) and adds $1000\, F_{\mathrm{rel}}\, D$ μg to A_g.

Table 2: Parameters: posterior medians of Notario *et al.* [6] (fitted to [4]), the scale factors fitted here, and derived quantities.

	Mitragynine		7-hydroxymitragynine		
k_a	1.99	h^{-1}	F_m	0.65	–
CL	17.85	L/h	CL_m	106.29	L/h
V_c	115.71	L	V_{cm}	50.61	L
Q	10.76	L/h	Q_m	8.52	L/h
V_p	355.64	L	V_{pm}	71.76	L
Fitted here (leaf powder)					
F_{rel}	0.277	–	s_{OH}	1.85	–
Derived					
absorption $t_{1/2}$	0.35	h			
α ($t_{1/2,\alpha}$)	0.2595	h^{-1} (2.67 h)	α_m ($t_{1/2}$)	2.2778	h^{-1} (0.30 h)
β ($t_{1/2,\beta}$)	0.0180	h^{-1} (38.54 h)	β_m ($t_{1/2}$)	0.1095	h^{-1} (6.33 h)

The MG disposition rate constants follow from $k_{10} = CL/V_c$, $k_{12} = Q/V_c$, $k_{21} = Q/V_p$ as $\alpha, \beta = \frac{1}{2}\left[(k_{10}+k_{12}+k_{21}) \pm \sqrt{(k_{10}+k_{12}+k_{21})^2 - 4k_{10}k_{21}}\right]$, and analogously for 7-OH. The resulting MG distribution half-life of 2.67 h and terminal half-life of 38.54 h agree with the "fast fall for about 6 h, then slow tail" pattern and with the long half-lives reported by studies that sampled for 5–10 days (Table 1). 7-OH's own disposition is fast (0.30 h and 6.33 h), so its plasma level is *formation-rate-limited*: after the first few hours it follows the MG curve with a roughly constant ratio. The short 7-OH half-lives reported clinically (1.7–5.7 h) reflect the fast distribution phase of the parent, as the metabolite falls below the limit of quantification before the slow tail is reached.

3.2 Direct 7-OH in the powder

The users' product contains 0.001 % 7-OH (0.01 mg per gram) against 16 mg MG per gram. Even assuming complete bioavailability of ingested 7-OH, its contribution to 7-OH exposure (same clearance CL_m) is at most $0.01/(s_{\mathrm{OH}}\, F_{\mathrm{rel}}\, F_m \cdot 16) = 0.19\,\%$ of the 7-OH formed from MG. It is therefore neglected.

3.3 Analytical solution and superposition

The system $\dot{\mathbf{A}} = \mathbf{K}\mathbf{A}$ is linear and time-invariant, and $\mathbf{K}$ is lower block-triangular with five distinct real eigenvalues $-\lambda_k \in \{-k_a, -\alpha, -\beta, -\alpha_m, -\beta_m\}$. The response to a unit dose is therefore a sum of exponentials,

$$h_{\mathrm{MG}}(\tau) = \sum_{k=1}^{5} a_k e^{-\lambda_k \tau}, \qquad h_{\mathrm{OH}}(\tau) = \sum_{k=1}^{5} b_k e^{-\lambda_k \tau}, \qquad \tau \geq 0, \tag{4}$$

obtained from the eigen-decomposition $\mathbf{K} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$: $\mathbf{A}(\tau) = \mathbf{V} e^{\mathbf{\Lambda}\tau}\mathbf{V}^{-1}\mathbf{A}(0)$. Here h is the response to 1 mg of MG entering the gut. For a dosing history $\{(t_i, g_i)\}$ (grams of powder at times t_i, i.e. $D_i = 16\, g_i$ mg MG) the concentrations are exact superpositions,

$$C_{\mathrm{MG}}(t) = F_{\mathrm{rel}} \sum_{i:\, t_i \leq t} D_i\, h_{\mathrm{MG}}(t - t_i), \qquad C_{\mathrm{OH}}(t) = s_{\mathrm{OH}} F_{\mathrm{rel}} \sum_{i:\, t_i \leq t} D_i\, h_{\mathrm{OH}}(t - t_i). \tag{5}$$

No numerical integration is needed in the app; the coefficients (per gram of the users' powder, already including F_{rel}, f_{MG} and s_{OH} for c_k below) are listed in Table 11. The coefficients of each analyte sum to zero, so $h(0) = 0$.

4 Calibration and validation

4.1 Procedure

The Notario parameters reproduce the tea study they were fitted to (model vs. observed [4]: MG $C_{\max}$ 151.9 vs. 159 ng/mL, $t_{\max}$ 1.20 vs. 0.84 h; 7-OH $C_{\max}$ 14.5 vs. 12.8 ng/mL, $t_{\max}$ 1.85 vs. 1.77 h). For leaf-powder capsules, exposure per mg MG is lower and the 7-OH:MG ratio is higher (Table 1). We therefore estimated a relative apparent bioavailability F_{rel} (scaling both analytes) and a metabolite scale s_{OH} (scaling 7-OH only) by least squares on the log scale, using the 16 single-dose mean values of Huestis *et al.* [1] ($C_{\max}$ and $\mathrm{AUC}_{0-\infty}$

of both analytes at four doses). The time-course parameters were not refitted. F_{rel} absorbs everything that makes the capsule cohort's exposure per mg lower than the tea cohort's (formulation, population, assay), and s_{OH} absorbs the higher metabolite ratio. Neither has a mechanistic interpretation.

Result: $F_{\mathrm{rel}} = 0.277$, $s_{\mathrm{OH}} = 1.85$, root-mean-square log error 0.382 (a typical deviation of $\times\, e^{0.382}$). The residual pattern is systematic (Table 3). (i) Peaks are 17–30 % lower than observed at every dose, i.e. the tea-derived time course is slightly flatter than the capsule data. (ii) AUC is over-predicted at low doses (about 2× at 6.65 mg) and slightly under-predicted at 53.2 mg, because observed AUC grows more than proportionally with dose while a linear model cannot. Both deviations are well within the between-subject range of the data.

Table 3: Calibration: single dose, observed (means, Huestis 2024 [1]) vs. model. $C_{\max}$ in ng/mL; $\mathrm{AUC}_{0-\infty}$ in ng h/mL.

	MG $C_{\max}$		MG AUC		7-OH $C_{\max}$		7-OH AUC	
mg MG	obs	model	obs	model	obs	model	obs	model
6.65	17.1	11.9	52.8	103	3.6	2.1	11.2	21
13.3	32.1	23.7	136	207	6.6	4.2	28.8	42
26.6	57	47.5	364	413	10.8	8.4	58.2	83
53.2	125	95.0	908	827	22.7	16.8	136	167

4.2 Validation against repeated dosing

The calibrated model was then used to predict day 15 of once-daily dosing, data not used in the fit (Table 4, Figure 2b). Peaks and AUCs are predicted within a factor of about 2 (the largest deviation is the 7-OH AUC at the lowest dose), and MG troughs within 20 % at the three lower doses. At 53.2 mg the trough is under-predicted (12.4 vs. 20.5 ng/mL), consistent with the longer half-life observed at high doses. The model reproduces the modest observed accumulation: the steady-state peak of the combined index is 1.145 times the single-dose peak (observed MG $C_{\max}$ ratio 1.1–1.3).

Table 4: Validation: day 15 of daily dosing, observed (means [1]) vs. model. Trough = concentration 24 h after the 15th dose.

	MG $C_{\max,ss}$		MG trough		MG AUC_{τ}		7-OH $C_{\max,ss}$		7-OH AUC_{τ}	
mg MG	obs	model	obs	model	obs	model	obs	model	obs	model
6.65	21.4	13.4	1.4	1.55	85.1	103	3.2	2.4	10.1	21
13.3	33.5	26.8	2.74	3.09	175	207	5.9	4.8	28	42
26.6	74.6	53.5	7.7	6.19	457	413	11.3	9.6	67.8	83
53.2	143	107.0	20.5	12.37	958	826	21.8	19.3	133	167

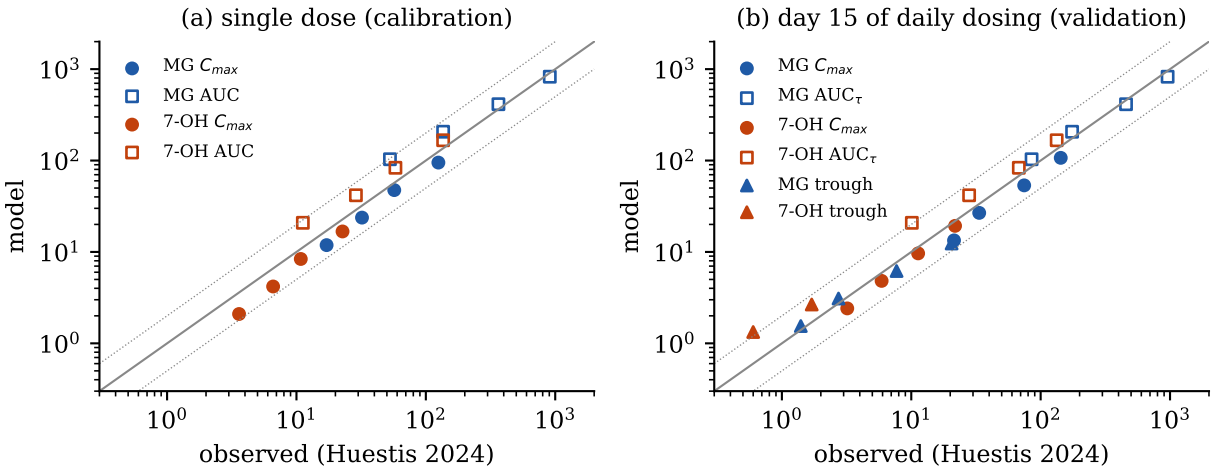


Figure 2: Observed vs. model. (a) Single doses (calibration data). (b) Day 15 of daily dosing (validation, not fitted). Solid line: identity; dotted lines: factor 2.

5 A combined activity index

The app shows one combined line rather than two concentrations. We define a potency-weighted MOR-equivalent concentration

$$E(t) = \frac{C_{\mathrm{MG}}(t)}{M_{\mathrm{MG}}} + w\,\frac{C_{\mathrm{OH}}(t)}{M_{\mathrm{OH}}} \quad [\mathrm{nM}], \qquad w = \frac{EC_{50,\mathrm{MG}}}{EC_{50,\mathrm{OH}}} = \frac{339}{34.5} = 9.83, \tag{6}$$

with molar masses $M_{\mathrm{MG}} = 398.5$ and $M_{\mathrm{OH}} = 414.5\,\mathrm{g/mol}$ (concentrations in ng/mL $\times 1000/M$ give nM). E is the MG concentration that would give the same receptor stimulation in the low-occupancy (linear) range. Underlying assumptions: (i) equal unbound fractions (>90 % bound for both [22]); (ii) equal access to the receptor compartment (brain distribution kinetics are not modelled); (iii) potency ratio from one *in vitro* assay [8]; (iv) the difference in $E_{\max}$ (34 vs. 47 %) and receptor saturation are ignored; (v) non-opioid targets are ignored.

With $w = 9.83$, 7-OH accounts for 66 % of the area under E after a single dose (63 % at the peak), in line with the finding that formed 7-OH mediates much of MG's opioid effect [9]. Because 7-OH follows the MG curve closely (formation-limited), the *shape* of E is insensitive to w (Figure 7a): it differs from MG alone mainly by a slightly later peak.

6 Relative scale shown to the user

Absolute concentrations are not shown. The displayed level is

$$L(t) = 100\,\frac{E(t)}{R \cdot G_0} \quad [\%], \tag{7}$$

where G_0 is the user's plan baseline (g/day, field `baselineG`) and $R = 75.55\,\mathrm{nM}$ per g/day is the steady-state *daily mean* of E when 1 g/day is taken. Because the model is linear, the steady-state mean depends only on the daily amount, not on how it is split ($\bar{E} = \frac{1}{24}\,G_0 \int_0^\infty h_E\,d\tau = \frac{G_0}{24}\sum_k c_k/\lambda_k$). Hence **100 % is the average level at the start of the taper**, and a plan that halves the daily amount brings the daily mean to 50 %. Peaks rise above 100 %: a single baseline-sized dose from a drug-free state peaks at 248 %, and taking the whole baseline once daily at steady state gives peaks of 284 % (accumulation factor 1.145). Splitting the amount into several doses gives lower peaks (Figure 4b).

Table 5: Model time course after a single dose, as % of each curve's own peak.

hours after dose	1	2	4	6	8	12	24	48	72	168
MG	99	91	58	37	24	12	5	3	2	0
7-OH	80	100	71	46	31	16	6	3	2	0
combined E	88	98	67	43	29	15	6	3	2	0

After stopping from steady state (baseline taken once daily), L falls to 37.5 % one day after the last dose, 23.3 % after 2 days, 15.1 % after 3 days, 2.7 % after 7 days and 0.1 % after 14 days (measured from the last dose, as % of the baseline daily mean). A dose 7 days old contributes 0.38 % of its peak and a dose 14 days old 0.02 %, so the app only needs the last 14 days of doses.

7 Amount remaining in the body

Alongside the relative level, the app shows how many milligrams of each analyte are still in the body. The model parameters are *apparent* (divided by the unknown absolute oral bioavailability F; no intravenous human data exist), so true amounts cannot be computed. We therefore report the *dose-equivalent amount not yet eliminated*, i.e. the amounts the model holds when the whole ingested dose is treated as absorbed (F_{rel} not applied):

$$M_{\mathrm{MG}}(t) = \sum_i D_i\,\big[A_1 + A_2\big]_{\mathrm{unit}}(t - t_i), \qquad M_{\mathrm{OH}}(t) = s_{\mathrm{OH}} \sum_i D_i\,\big[A_3 + A_4\big]_{\mathrm{unit}}(t - t_i), \tag{8}$$

where $[\cdot]_{\mathrm{unit}}$ is the response to 1 mg in the gut. Unabsorbed drug in the gut is not counted. This is an *upper estimate* of the amount actually in the body, and it is the natural reading of "how much of what I took is still in me". Because most MG resides in the peripheral compartment ($V_p \approx 3V_c$), M_{MG} changes much more

smoothly than plasma concentration. Example (70 kg adult, the population of the calibration studies; 6 g/day in three doses, day 14): M_{MG} ranges 87–121 mg and M_{OH} 3.8–6.8 mg over the day, against 96 mg MG ingested per day. The model is not scaled for body weight. Coefficients are listed in Table 6.

Table 6: Coefficients for the amounts still in the body, $M(\tau) = \sum_k d_k e^{-\lambda_k \tau}$, in mg per gram of the users' powder.

k	λ_k (h^{-1})	d_k MG (mg/g)	d_k 7-OH (mg/g)
1	0.017984	+7.03624	+0.19781
2	0.109472	0	+1.48270
3	0.259527	+10.38122	-0.29434
4	1.990000	-17.41746	-10.66058
5	2.277782	0	+9.27441

8 Model behaviour

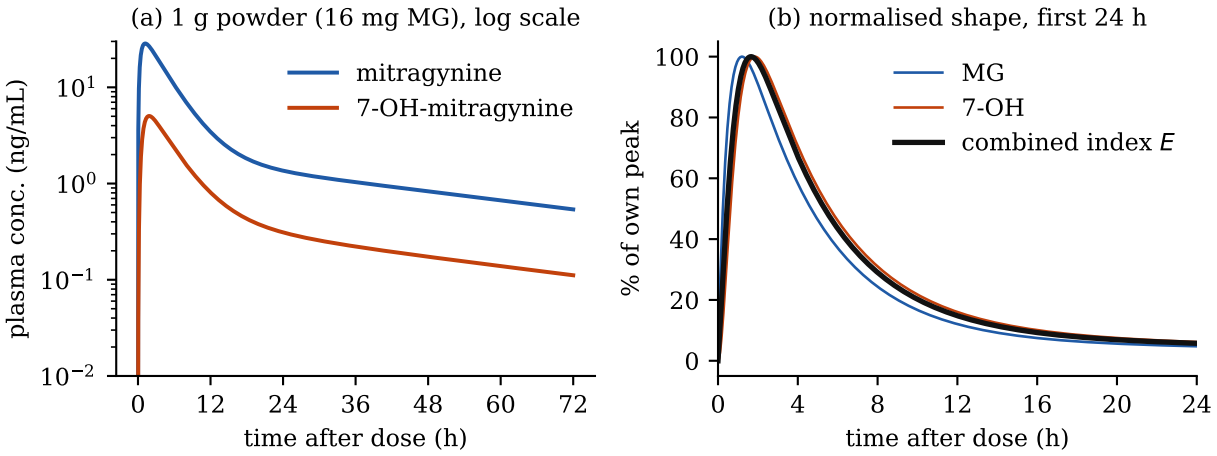


Figure 3: Single dose of 1 g powder (16 mg MG). (a) Absolute model concentrations, log scale: fast distribution phase, then a terminal phase with $t_{1/2} \approx 38.54$ h; 7-OH runs parallel to MG. (b) Normalised shapes over 24 h.

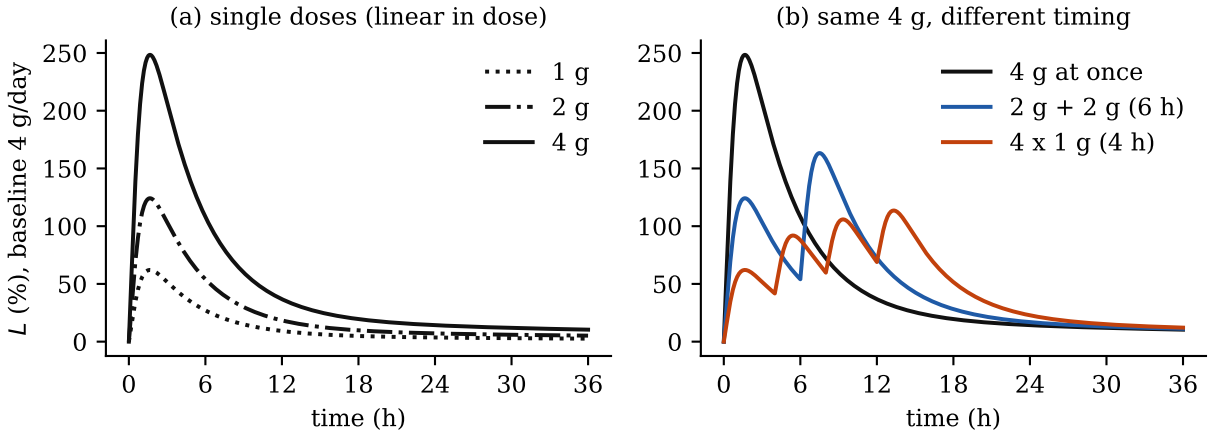


Figure 4: Dose and timing responsiveness, shown as L for a user with baseline $G_0 = 4$ g/day. (a) Peaks scale linearly with dose. (b) The same daily amount split into more doses gives a lower, broader curve.

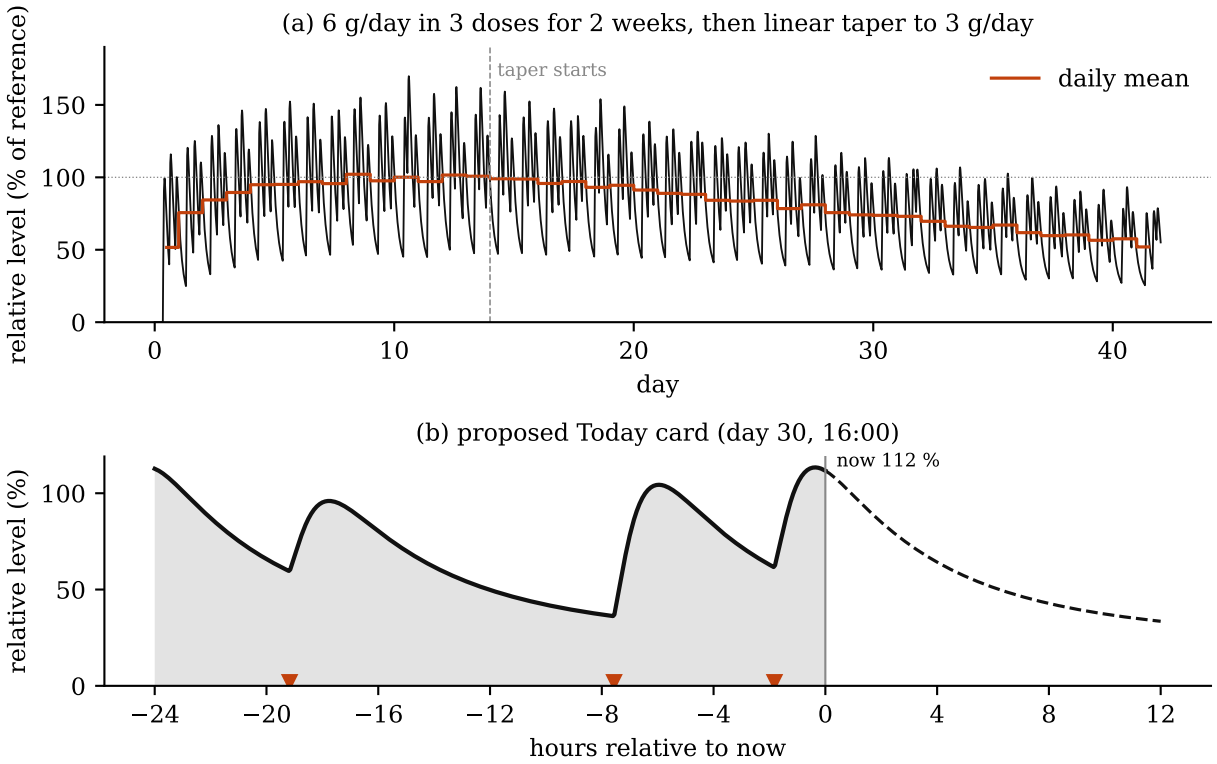


Figure 5: Simulated user (illustrative, random dose times): baseline 6 g/day in three doses for two weeks, then a linear taper to 3 g/day. (a) $L(t)$ over six weeks and its daily mean, which starts at 100 % and ends near 50 %. (b) The proposed "Today" card: last 24 h (solid), the moment "now", and the projection if no further dose is taken (dashed); triangles mark doses.

9 Regular and heavy users (version 2)

9.1 Question

Every user of the app has taken kratom leaf powder daily for more than a year (up to about four years), several at peak amounts above 30 g/day. Version 1 was calibrated on Huestis *et al.* [1], whose participants were kratom-naive or had not used kratom for ≥12 months, were non-smokers without a history of drug use, took at most 4 g/day for 15 days and were fasted. At 1.6 % MG, 10–30 g/day is 160–480 mg MG per day, three to nine times the highest repeated dose ever studied (53.2 mg). We asked: (i) do long-term daily users handle mitragynine differently, and (ii) is there any PK information for the amounts our users take? The user also proposed choosing model parameters per user from the logged intake; whether the data support this is part of (i)–(ii).

9.2 Search

Searches were run on 12 November 2026.

- **PubMed** (E-utilities): kratom/mitragynine/7-OH/speciociliatine/ketum/biak combined with PK, plasma, serum, blood, urine, hair, oral fluid, half-life, clearance; with chronic, regular, long-term, daily, dependent, heavy, users; with CYP induction, PXR, steady state, multiple dose; case reports with measured concentrations; Malaysian, Thai and Indonesian user studies; author searches for the main clinical groups. About 600 unique records.
- **Europe PMC** full-text search (body text of open-access papers mentioning ng/mL together with daily use), which also covers preprints.
- **OpenAlex** forward citations of the seven key human PK papers (293 citing works), keyword searches, and a filter for preprints (medRxiv, bioRxiv, Research Square, OSF, ChemRxiv, Preprints.org, SSRN).
- **ClinicalTrials.gov**: all registered kratom/mitragynine studies (15) and posted results.
- Full texts were read for every study with human concentration data; five were obtained through institutional access. The supplement of Reissig 2026 [11] (Supplemental Digital Content 3) was also read:

it gives plasma curves of the minor alkaloids paynantheine, speciogynine and speciociliatine (figures only), pupil diameter, and subjective-effect scores (drug liking and "high" differed from placebo only at 12 g). It has no further mitragynine or 7-OH concentrations, so it does not change the comparison below. One case letter with measured plasma levels (Rianprakaisang 2023) could not be obtained.

Studies were kept if they reported mitragynine (or 7-OH) concentrations in humans by a specific chromatographic assay together with the dose or the use pattern. Animal, *in vitro*, and post-mortem studies were excluded from the quantitative analysis.

9.3 What exists

Table 7: Human data relevant to regular and heavy users. Only the first four can be compared quantitatively with the model.

Study	Population	Use for the model
Reissig 2026 [11]	single doses 1–12 g leaf, fed, 48 h sampling; not dependent	highest doses with PK; linearity, Table 8
Trakulsrichai 2015 [5]	9 Thai men, 0.6–5 y of use, 1–9 leaves/day; 7 days on a fixed tea dose	peaks at steady state, Table 9
Smith 2024 [12]	10 US regular users, 2–14 y of use, usual morning dose	post-dose levels, 7-OH ratio, Table 9
Kanumuri 2026 [13]	4 US regular users, usual dose, serial sampling	pre-dose levels, half-lives, Table 9
Wahyono 2025 [15]	64 Indonesian users, mostly 2–5 y, often >21 leaves/day; serum by ELISA	not usable: mean 16 µg/mL, about 100× above every LC-MS value, so the antibody also detects metabolites; its "no relation to intake" is uninformative
Ang 2026 [16]	Malaysian dependent users, spot urine (58–18 000 ng/mL)	not usable: no amounts or timing
Leong Abdullah 2020 [17]	100 Malaysian regular users, estimated 434 mg MG/day	intake only (confirms that 400+ mg/day users exist); no blood levels
Kedzierski 2023 [18]	60 drivers, blood 10.5–960 (median 58) ng/mL	plausibility only (no doses); same order as the regular-user levels in Table 9
Peřan 2023 [19], Holton 2024 [20]	Czech poisonings; one user at ∼100 g/day for 3 y	no concentrations
NCT06089980 [21]	regular users stopping under observation, serial blood	the decisive study for the decline after stopping; recruiting, completion 2028

No preprint and no registered trial with posted results adds human PK data. In short: **no time-resolved PK data exist for long-term users taking more than about 10–12 g/day.** The closest evidence is single doses up to 12 g in people who do not use daily, and regular users at their usual (smaller) doses.

9.4 Comparison with the model

The version-1 model (unchanged, $F_{\mathrm{rel}} = 0.277$ from fasted capsules) was used to predict each study.

Table 8: Reissig 2026 [11]: single doses (fed), observed means vs. model. $C_{\max}$ in ng/mL, AUC_{0-48} in ng h/mL. Last column: observed/model ratio of the MG AUC.

		MG $C_{\max}$		MG AUC		7-OH $C_{\max}$		7-OH AUC		
g	mg MG	obs	model	obs	model	obs	model	obs	model	ratio
1	10	41.7	18	305.4	130	6.7	3.2	53.5	26	2.35
3	31	118	55	784.8	391	17.8	9.7	132.9	78	2.01
8	82	197.5	147	1942	1041	34	25.9	412.6	209	1.87
10	103	318.1	184	3254	1301	48.4	32.4	459.1	261	2.50
12	123	378.5	220	3271	1562	58.4	38.9	574.3	313	2.09

Table 9: Regular users vs. model. Daily frequency is unknown in Smith and Kanumuri, so the model is shown for 3, 4 and 6 equal doses a day (doses spread over the waking day; overnight gap before the morning dose).

Trakulsrichai 2015: day-8 peak (ng/mL) observed 18.5–105			
mg/day	test dose	capsule F	tea F
6.25	12.5	24	85
6.25	6.25	13	45
9.96	19.92	38	136
9.96	9.96	20	72
11.5	23	44	157

Smith 2024: 54 mg morning dose observed MG 176, 7-OH 34 (ng/mL)			
doses/day	MG	7-OH	ratio
3	143	27	0.19
4	161	31	0.19
6	198	39	0.19

Kanumuri 2026: concentration before the usual dose (ng/mL)					
participant	usual dose (mg MG)	observed	model 3/day	4/day	6/day
KIDN01	32.3	26.9	29	41	66
KIJC02	26.6	58.5	24	34	54
KIDL03	54.1	222.4	49	69	110
KIDM04	29.8	107.7	27	38	61

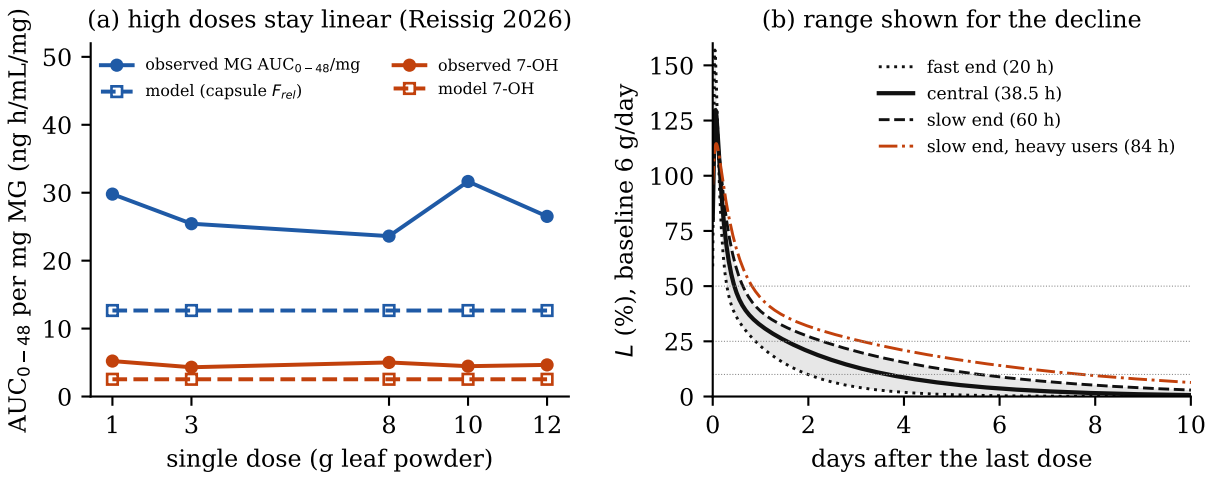


Figure 6: (a) Exposure per mg of MG after single doses of 1–12 g (Reissig 2026): flat, i.e. linear, and about twice the model's. (b) Decline after stopping 6 g/day (taken as the baseline), with each variant of §9.5 expressed against its own reference. The band between the fast and slow ends is what the app shows.

Findings.

1. **Linear up to 12 g.** MG AUC_{0-48} per mg was 23.6–31.7 ng h/mL/mg over a 12-fold dose range with no trend (Figure 6a); the authors describe Cmax as slightly *less* than dose-proportional, with absorption possibly limited by the number of capsules at 10–12 g. The supra-proportionality seen in [1, 2] (up to 1.4×) was not reproduced. The linear model is supported up to single doses of about 125 mg MG.

2. **Absolute exposure is about 2× the model's.** Observed/model geometric-mean ratios in Reissig were 1.81 (MG C_{max}), 2.15 (MG AUC), 1.62 and 1.86 (7-OH), with the same factor at every dose. Trakulsrichai's peaks lie between the model with capsule and with tea absorption; Smith's levels are matched by 4–6 doses a day; two of Kanumuri's four pre-dose levels are 2–4× the model's. Fed administration, loose powder or tea and population all plausibly raise absorption compared with fasted capsules. **This factor does not change anything the app shows:** the percentage divides E by the steady-state reference R, which scales with the same factor; "kratom by effect" divides by the model's own 1 g peak; and the milligrams are dose-equivalent amounts that assume complete absorption (§7). The constants were therefore left unchanged, as was the SQL copy.

3. **The 7-OH share holds.** In regular users the 7-OH/MG ratio after the morning dose was 0.19 (34/176 [12]); the model gives 0.19 at steady state (Table 9). The potency-weighted combination is

unchanged. Note that 7-OH degrades in unstabilised plasma samples [13], so earlier 7-OH values may be underestimates.

4. **No sign that long-term use speeds up elimination.** *In vitro*, kratom alkaloids activate PXR and induce CYP3A4 and CYP1A2 [14], which could in principle shorten MG's half-life in daily users. In humans, half-lives in regular users (17–35 h [13]; 23 ± 16 h [5]) are within the range of non-users with similar sampling (16–22 h over 48 h [11]), and the multiple-dose half-life in [1] was longer, not shorter, than the single-dose one.
5. **Dose classes are not supported.** The apparent rise of half-life with dose in [1] is attributed by the same authors to assay sensitivity (higher doses stay measurable longer) [2], and Reissig's half-lives do not rise with dose. No data show that a 20 g/day user clears MG differently from a 5 g/day user, so per-class parameters would be invented. Intake is used only to widen the uncertainty range (§9.5).
6. **Other alkaloids.** In regular users speciociliatine reached plasma levels similar to MG (188 vs. 176 ng/mL [12]) and it is a more potent MOR partial agonist. It remains outside the model (user's decision to keep two components); this is listed as a limitation.

9.5 Range shown for the decline

What the data leave open is how fast the level falls, which drives the "when will it drop" projection. Reported MG terminal half-lives are 16–23 h with 24–48 h sampling or in regular users [5, 13, 11], and 43–68 h with 5–10 days of sampling and after repeated dosing [1, 3]. The central model has 38.5 h. Two variants span this range: every disposition clearance (CL, Q, CL_m, Q_m) is multiplied by a factor s, which multiplies all disposition rate constants by s (absorption is unchanged) and so divides the terminal half-life by s:

- **fast end**: $t_{1/2} = 20$ h ($s = 1.93$);
- **slow end**: $t_{1/2} = 60$ h ($s = 0.64$);
- **slow end for heavy users**: $t_{1/2} = 84$ h ($s = 0.46$), used when the user's average over the last 14 days (or since their first dose, if more recent) exceeds 12 g/day. 12 g is the largest amount with any human PK data (Reissig's 12 g single dose; regular users in [12, 13] took up to about 11 g at once). Above it, the only hint is the possible enzyme saturation raised in [2], so the slow end is extended by the largest supra-proportionality reported ($\times 1.4$).

Each variant is shown against *its own* steady-state reference R_s (the analogue of R), and without a plan against its own 1 g peak. So "50 %" always means half of that variant's usual level: on a steady routine all variants agree, and they spread out only when intake changes. Table 10 shows the resulting range after stopping; the variants use a 21-day dose window, which keeps the truncated tail below 1 % even for the slowest one (a 14-day window would drop 0.45 % of a single dose's peak).

Table 10: Time from the last dose until the level falls below 50, 25 and 10 % of the baseline level, after 30 days of 6 g/day in three doses (= the baseline).

	terminal $t_{1/2}$ (h)	below 50 %	below 25 %	below 10 %
fast end	20	7 h	21 h	48 h
central	39	11 h	37 h	3.7 d
slow end	60	15 h	2.3 d	5.6 d
slow end, heavy users	84	20 h	3.1 d	7.7 d

In the app. The *V těle teď* card shows the central projection as before, a light band from the fast to the slow end on the chart, and under each "below X %" line the earliest and latest time from the two ends. Users above 12 g/day see a note that their amount is beyond what has been measured and that the slow end is extended. Morning recaps and milestone notifications (SQL) still use the central curve.

10 Uncertainty and limitations

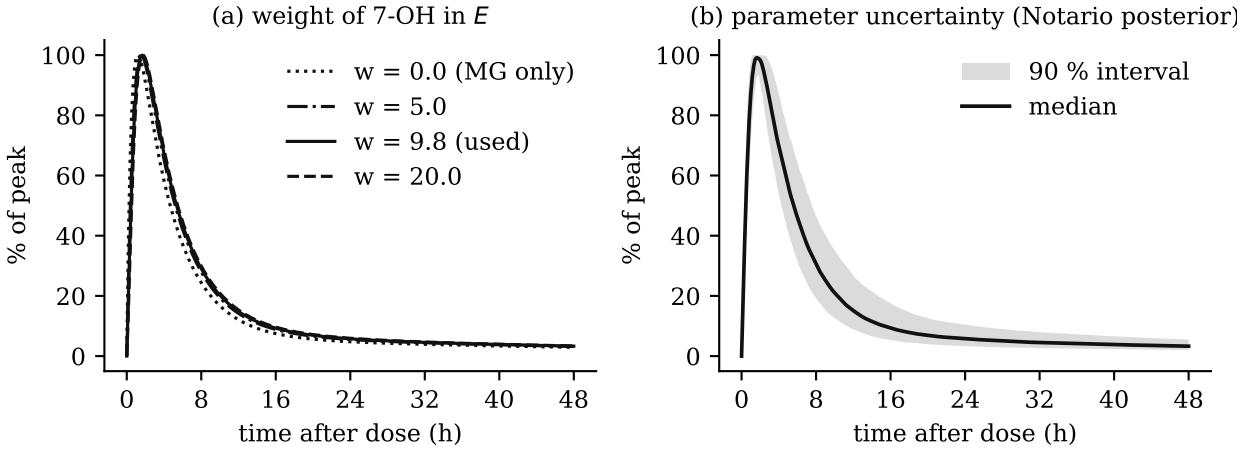


Figure 7: (a) Shape of E for different 7-OH weights w. (b) Uncertainty of the population-average shape from the parameter posterior (log-normal approximation of [6], 400 draws). At 12 h the level is 15 % of peak (90 % interval 9–27 %), at 24 h 6 % (3–10 %).

1. **Population average, not the individual.** Between-subject variability in exposure is 5–10-fold [1] and is not represented. Figure 7b shows only the uncertainty of the *average* curve. Normalising to the user's own baseline removes the overall scale, but not individual differences in shape (e.g. CYP3A4 activity).
2. **Linearity.** AUC rose up to ∼1.4× faster than dose in [1, 2] but not in single doses up to 12 g [11] (§9.4). Linearity is untested for repeated daily amounts above about 4 g and for single doses above 12 g.
3. **Calibration source.** One capsule study, with fasting. Under real-world conditions absolute exposure is about twice the model's (§9.4); this does not affect the displayed quantities. Food delays absorption (Reissig: $t_{\max}$ 2–5 h when fed) and can cause secondary peaks [1].
4. **Long-term and heavy use.** Regular users at their usual doses agree with the model (§9.4), but there are no time-resolved data for long-term users taking more than about 12 g/day, nor for the decline after such users stop [21]. The projection is therefore shown as a range (§9.5), wider for heavy users.
5. **Product composition.** 1.6 % MG is taken as fixed; real batches vary. Other alkaloids are ignored, although in regular users speciociliatine reaches plasma levels similar to MG [12] and is a more potent MOR partial agonist.
6. **Interactions.** CYP3A4 inhibitors (e.g. itraconazole, clarithromycin, grapefruit) lower 7-OH and raise MG [4]; inducers (e.g. rifampicin, St John's wort) are expected to do the opposite. Not modelled.
7. **Concentration is not effect.** Tolerance, receptor adaptation and withdrawal are not modelled; withdrawal symptoms do not map onto a plasma level. A low displayed level does not mean that a dose is needed.
8. **Not applicable** to extracts (higher exposure per mg [2]) or to semi-synthetic 7-OH products.

11 Implementation specification

Table 11: Closed-form coefficients for $h(\tau) = \sum_k \mathrm{coef}_k\, e^{-\lambda_k \tau}$ (τ in h). a_k: MG in ng/mL per mg MG reaching the gut (multiply by $16\,F_{\mathrm{rel}}$ per gram); b_k: 7-OH (unscaled) likewise; c_k: combined index E in nM per gram of the users' powder (all factors included).

k	λ_k (h^{-1})	$t_{1/2,k}$ (h)	a_k	b_k	c_k
1	0.017984	38.54	+0.443066	+0.049497	+14.56008
2	0.109472	6.33	0	+0.051653	+10.04373
3	0.259527	2.67	+9.433512	+1.005442	+300.59118
4	1.990000	0.35	-9.876578	-7.829166	-1632.38407
5	2.277782	0.30	0	+6.722574	+1307.18908

1. **Module** `src/lib/pk.ts`: constants λ_k, c_k, R and the amount coefficients of Table 6 (copied from `ts_constants.json`); a pure function `level(doses, t, baselineG)` implementing $L(t) = 100 \sum_i g_i \sum_k c_k e^{-\lambda_k (t-t_i)} / R$ (with $R = 75.55$) over doses with $0 \le t - t_i \le 14$ d, and a helper that returns a series on a 10-min grid for charting.
2. **Data.** The user's own doses for the last 14 days (`takenAt`, `grams`) and the plan's `baselineG`. Runs entirely in the browser; no database change. Other users' curves are never computed (their dose times are private).
3. **Today card** (below the list of today's doses). Milligrams of MG and 7-OH still in the body (§7), current L (rounded to 5 %; it can exceed 100 % shortly after a dose), a 24 h history, dose markers and a dashed 12 h projection with no further doses. Without an active plan there is no baseline, so the card shows the amounts and the curve without a percentage scale. Doses logged at a later time today appear when their time comes.
4. **Tests** (`pk.test.ts`): (i) the sums of exponentials agree with a numerical ODE solution; (ii) the single-dose and day-15 values in Tables 3 and 4; (iii) superposition and linearity; (iv) the daily mean of L is 100 % at steady state on the baseline amount, however it is split.
5. **Wording.** Always labelled as a rough population-average estimate, not a measurement, and with no advice derived from it.
6. **Range (version 2).** `PK_VARIANTS` (fast, slow, slowHeavy: λ_k, c_k, R_s from `ts_constants.json`), `pkAverageGramsPerDay` (14-day average, or since the first dose) and `pkRangeVariants` (slow end switches above 12 g/day). The chart band and the earliest/latest projection times use these variants with a 21-day dose window. Tests check the variants against the scaled ODEs, their terminal half-lives, R_s, and the 50 % times of Table 10.

Reproducibility

All numbers, tables and figures in this report are generated by `docs/pk-model/model.py` (NumPy, SciPy, Matplotlib), which writes `numbers.tex`, `numbers.json` and `fig/*.pdf`.

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